

Trading Like a Bayesian: Confidence-Weighted Time-Series Momentum, and Why It Does Not Beat the Original

Gator Quant Hacks 2026 · Systematic Trading Track · [Team member names] · Code:
github.com/gabejsto-ops/GQH (one command: `python run_all.py --final`)

1. Summary

We start from conservatism: investors update toward news by less than Bayes' rule implies, so prices drift after information and time-series momentum (TSMOM) captures the drift. We tried to use Bayesian reasoning to improve TSMOM on 23 liquid ETFs (2008–2026): first by de-risking only on noisy volatility, then by sizing each position by the posterior probability that its trend is real. We pre-registered every hypothesis in git, ran **15 strategy variants and 3 mechanism tests**, held out the last two years and evaluated them once. A naive backtest looks strong: an out-of-sample Sharpe of **1.08** net of costs. Our own audit takes it apart. Measured over cash, it is **0.21** (95% CI -0.72 to 1.58). The Bayesian sizing is indistinguishable from standard TSMOM, and both barely beat simply holding the same vol-targeted portfolio. What survives is the known property of trend-following: a similar long-run Sharpe to long-only (~ 0.4) with protection in crashes.

2. Economic Hypothesis

Edge source (behavioral). In a Bayesian update, $new = old + k(news - old)$. Conservative investors use a gain below k (Edwards 1968; Barberis, Shleifer & Vishny 1998), so part of each repricing arrives late as drift. **Who is on the other side:** investors anchored to pre-news valuations, plus hedgers and policy makers who trade gradually for non-profit reasons. **Why it persists:** the bias is human, and trend-following suffers long whipsaw drawdowns that limit the capital competing it away. The base effect is documented across 58 futures by Moskowitz, Ooi & Pedersen (2012).

Our extensions (committed before any backtest, commit 06b9b69; Round 2 committed d0e5746 before running). *Round 1 (H1/H2):* high volatility mixes information shocks (large reliable news, large k , so trends should continue) with noise (small k , so moves reverse). Standard TSMOM cuts size in both; we predicted that cutting only in noisy, inefficient high-vol months would improve it. *Round 2 (H4):* a Bayesian bets in proportion to confidence. With a flat prior, $P(\text{drift} > 0) = \Phi(z)$, where z is the t-stat of the trailing mean return, so position = vol-target weight $\times (2\Phi(z) - 1)$. We predicted higher Sharpe and lower turnover than sign-based TSMOM. **Each hypothesis fails if** it does not beat S1 net of costs; the mechanism tests fail if the conditional difference is ≤ 0 or insignificant.

3. Data & Universe

Universe (fixed by rule): the most liquid US-listed ETF per distinct exposure with inception before mid-2007: 8 equity (SPY, QQQ, IWM, EFA, EEM, EWJ, EWZ, FXI), 6 bond (SHY, IEF, TLT, LQD, HYG, TIP), 5 commodity (GLD, SLV, DBC, USO, DBA), 3 currency (UUP, FXE, FXY), 1 real estate (VNQ). **Sources:** Yahoo Finance daily bars via `yfinance`, split- and dividend-adjusted (`auto_adjust`); risk-free rate = 3-month T-bill (FRED DTB3); factors from the Kenneth French Data Library. We chose public data over the Webull API so judges can reproduce results without credentials. **Sample:** 2007-04-11 (HYG launch) to 2026-10-02; signals start 2008-05 after a one-year warm-up. **Out-of-sample:** the last 2 years (2024-10-03 onward), since 20% of 19.5 years is longer than the 2-year cap. The loader drops those rows unless `--final` is passed.

Data problems. *Survivorship:* all 23 still trade and were picked knowing they are liquid today; we also excluded UNG by name because we knew its roll losses. Both are hindsight. *Corporate actions:* adjusted prices; USO's 1-for-8 reverse split (2020-04-28) checked as correctly adjusted. *Missing data:* none in

sample. All four one-day moves above 25% were verified as genuine (EWZ 2008-10-13; USO 2020-03-09 and 2020-04-21; SLV 2026-01-30). Nothing is filled forward.

4. Methodology

Signal and timing. At each month-end close, signal = sign of the trailing L -day return. Trades fill at the **next day's open**; positions drift with prices until the next rebalance. **Sizing** (portfolio budget 10% vol split equally, per-asset cap $2\times$ equity, 60-day realized vol): **S0** sign only, $\pm 1/N$; **S1** vol-targeted (standard TSMOM); **S2** S1 but sized with 252-day vol in "information-shock" months (21-day vol above its trailing-year 75th percentile *and* 21-day efficiency ratio $|\text{net move}| / \text{path length} > 0.44$, twice the random-walk level $1/\sqrt{21}$); **S3** $S1 \times (2\Phi(z) - 1)$, the Bayesian sizing; **S4** S1 doubled in calm, efficient months (the low-vol mirror of S2). Thresholds come from theory, not fitting; S3 adds no free parameters. **Variants:** $L \in \{63, 126, 252\} \times 5$ sizings = **15 strategies**, all reported. After Round 2 we selected S3 at the pre-registered $L = 252$. The lookback was fixed in advance, but the choice of sizing rule depended on results, and we disclose that.

Costs. 5 bps per side on traded notional (commission-free broker; half-spreads are about 1 bp for the largest ETFs and several bps for thin ones such as FXY, DBA and UUP; plus open-auction slippage). Every result is also shown at **10 bps**. No borrow cost on shorts or financing on leverage (see §8). **Returns are measured in excess of cash:** $\sum_i w_i(r_i - r_f)$, the TSMOM-literature convention. We added this after our audit found the original engine credited T-bill carry (§5).

Tests. Mechanism tests on 4,500 asset-months (outcome = signal $\times$ next-month excess return / 60-day vol; Welch and month-clustered OLS): H1 shock vs noisy high-vol months; H3 continuous slope on efficiency; H5a calm efficient vs inefficient. Overfitting controls: Deflated Sharpe Ratio (Bailey & López de Prado 2014) over $N = 15$; 21-day block-bootstrap 95% CIs (2,000 draws); fills delayed by 1 and 4 extra days (lookahead check); a lookback sweep of 21–252 days (plateau check); a long-only baseline (S1's weights, always long); comparison at matched 10% vol; Fama-French 3 + momentum regression (HAC). The engine was verified on synthetic prices (zero weights give zero P&L; a single long position matches buy-and-hold exactly; an asset earning exactly r_f shows zero excess return).

5. Results

L = 252, 5 bps, over cash	In-sample 2008-06 → 2024-10			Out-of-sample 2024-10 → 2026-10		
	S3 (final)	S1	Long-only	S3 (final)	S1	Long-only
Excess return (ann.)	1.16%	1.55%	1.62%	0.71%	1.10%	1.94%
Volatility	3.08%	4.40%	5.40%	3.67%	4.90%	5.71%
Sharpe	0.39	0.37	0.33	0.21	0.25	0.36
Sharpe 95% CI	−0.09, 0.87	−0.10, 0.84	−0.19, 0.85	−0.72, 1.58	−0.69, 1.65	−0.45, 1.86
Sharpe at 10 bps	0.34	0.31	—	0.18	0.21	—
Max drawdown	−7.1%	−11.8%	−14.7%	−3.6%	−4.7%	−5.6%
Turnover ($\times/\text{yr}$)	3.2	5.1	2.1	2.6	3.9	1.9
Worst month	−2.7%	−3.4%	−5.8%	−2.5%	−3.2%	−2.6%
Skew (daily)	−0.67	−0.42	−0.95	−0.51	−0.41	−0.13
<i>Naive Sharpe (with carry)</i>	<i>0.49</i>	<i>0.45</i>	<i>0.56</i>	<i>1.08</i>	<i>1.06</i>	<i>1.22</i>

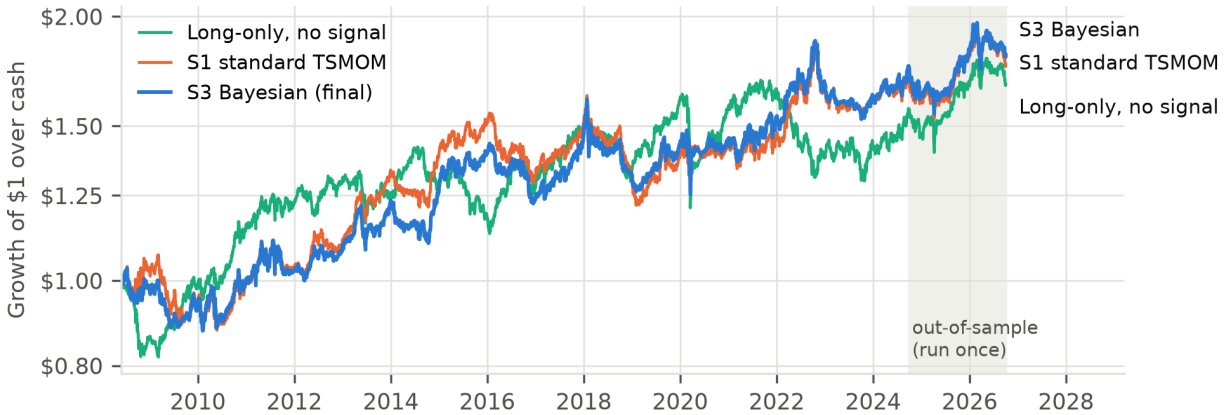


Figure 1. Growth of \$1 over cash, net of 5 bps, each strategy scaled to 10% vol using in-sample vol only. Shaded: out-of-sample, evaluated once.

The 15 variants (in-sample excess Sharpe, 5 bps; Deflated-Sharpe hurdle for N = 15 is 0.23)

Lookback	S0 sign	S1 vol target	S2 shock-aware (R1)	S3 Bayesian (R2)	S4 calm tilt (R2)
63 days	0.18	0.38	0.38	0.35	0.39
126 days	0.31	0.52	0.52	0.54	0.53
252 days (pre-reg.)	0.06	0.37	0.37	0.39	0.37

Best Deflated Sharpe: 0.89 (126-day S3/S4); none passes 0.95. Out-of-sample results for every variant are in the appendix.

What failed (the hypothesis log)

- **H2, S2 shock-aware de-risking: failed.** Only 2.3% of asset-months qualify as shocks, so $S2 \approx S1$ (Sharpe gap ≤ 0.004).
- **H1, shock months trend more: not significant.** 0.034 vs 0.009 ($n = 105$ vs 995), Welch $p = 0.56$, clustered $p = 0.65$.
- **H3, continuous version: failed.** Slope -0.02 , $p = 0.82$ ($n = 1,100$). Out of sample the slope is $+0.59$, $p = 0.006$; given the in-sample null, $n = 133$ and the number of tests run, we treat that as chance.
- **H5, calm efficient trends (S4): failed.** H5a 0.027 vs 0.014, $p = 0.39$; small Sharpe gains come with 30–50% more turnover and vanish at 10 bps.
- **H4, Bayesian sizing (S3): partly supported in-sample** (beats S1 at 126 and 252 days, not 63; turnover -37%), then **dismantled below**.

Tearing down our own headline

- **Cash carry (1.08 $\rightarrow$ 0.21).** The first engine credited bond-ETF interest but paid none on cash. With T-bills at about 4.5% and S3 about $0.8\times$ net long, 3.6 of its 4.5 points of out-of-sample return were the T-bill rate. The apparent improvement out of sample was an artifact of high rates.
- **Size, not skill.** S3 runs at lower vol, which flattered its turnover and drawdown. At matched 10% vol, turnover is $10.5\times$ vs $11.6\times$ (-10% , not -37%) and max drawdown -22% vs -25% in-sample, and the drawdown advantage is gone out of sample (-9.7% vs -9.4%). S3 – S1 Sharpe: $+0.02$ $[-0.14, 0.18]$ in-sample, -0.04 $[-0.26, 0.15]$ out of sample; correlation 0.95–0.98.
- **Not significant.** Every interval includes zero (in-sample $P(\text{Sharpe} \leq 0) = 0.055$, borderline); no Deflated Sharpe passes 0.95.

- **The signal barely beats holding.** Long-only with identical vol targeting earns 0.33 in-sample and *beats* both trend versions out of sample (0.36).
- **Known factors.** S3 in-sample loads on momentum ($\beta = 0.09$, $t = 19$) with a small market β of 0.035. Alpha is 0.9%/yr, $t = 1.26$ ($R^2 = 0.25$). Out of sample, market β rises to 0.12 ($t = 9.8$) and alpha is negative.
- **Regime dependence.** Bonds were 60% of in-sample excess P&L (sleeve Sharpe 0.52) and the worst sleeve out of sample (-0.93); out-of-sample gains came from commodities (1.32) and equities (0.66).

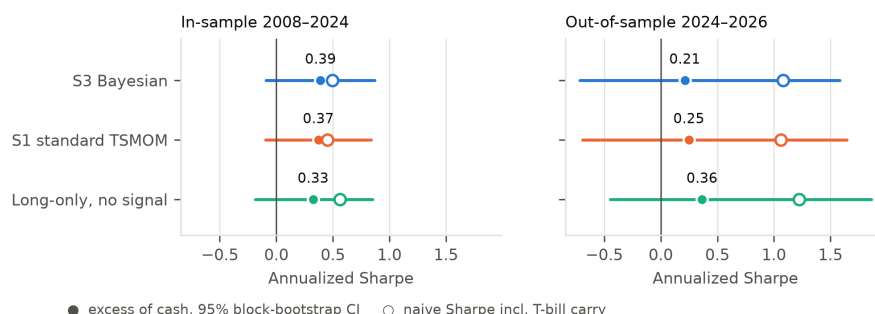


Figure 2. Excess Sharpe with 95% block-bootstrap intervals (filled) against the naive Sharpe including T-bill carry (hollow).

What survives. No lookahead: one and four days of extra fill delay lower S3 smoothly ($0.39 \rightarrow 0.37 \rightarrow 0.36$). A plateau: Sharpe 0.30–0.54 for 42–252-day lookbacks (Figure 3). Robust to doubled costs (-0.05). Positive in 12 of 17 calendar years. And the crash profile that separates trend from long-only (excess return, S3 / S1 / long-only): GFC 2008-09 to 2009-03 0.0% / $+2.4\%$ vs -5.9% ; COVID crash -3.0% / -2.3% vs -12.8% ; 2022 $+5.9\%$ / $+9.9\%$ vs -11.0% . The cost is rebounds: March–June 2009 -3.1% / -5.2% vs $+4.6\%$.

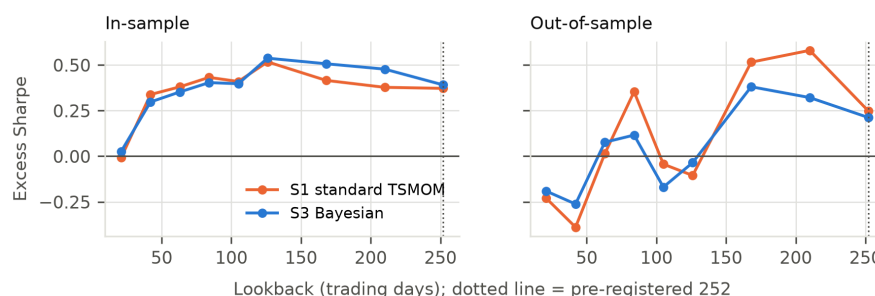


Figure 3. Excess Sharpe across lookbacks (diagnostic only, not used for selection). Smooth in-sample; noisy over 2 years out of sample.

6. Risk Management

Built into the backtest: per-asset vol targeting (10% portfolio budget / 23), a $2\times$ equity cap per asset, monthly rebalancing, 23 assets across 5 classes, and confidence scaling (average gross $0.80\times$, maximum $2.14\times$; average net $0.50\times$). **Observed risk (in-sample, S3):** daily 99% VaR -0.59% , CVaR -0.79% , worst day -1.42% (2022-11-10), longest drawdown 1,032 trading days. **Main weakness found:** vol targeting levers low-vol bonds. SHY reached 164% of equity and the bond sleeve 192% gross, so a rate shock is the concentrated risk. **Proposed rules for live trading** (stated in advance, deliberately *not* backtested to avoid a 16th trial): (1) gross cap of 100% of equity per asset class and $2.5\times$ in total; (2) if drawdown exceeds $2\times$ the target annual vol, halve the vol target until half the loss is recovered; (3) monitor 1-year rolling market β and cut the equity sleeve if $\beta > 0.3$; (4) retire the strategy if its 3-year excess Sharpe is below zero *and*

drawdown exceeds the backtest maximum (−7.1%). **Tail:** the documented weakness is sharp V-shaped reversals (2009, March 2020), where trend is short at the bottom; we do not hedge it.

7. Liquidity & Capacity

Positions are monthly trades in ETFs whose 2024 median daily dollar volume ranges from about \$10M (FXY) to billions (SPY, HYG, LQD). We model cost per trade as fixed 5 bps + square-root impact $\sigma_{\text{daily}} \cdot \sqrt{(\text{trade } \$ / \text{ADV } \$)}$, using the trailing 20-day ADV known before the trade. S3's in-sample excess Sharpe (0.44 gross of costs) falls to **0.38 at \$1M, 0.33 at \$100M, 0.20 at \$1B and −0.20 at \$10B**; out of sample 0.21 → 0.16 at \$100M. At \$1B the median trade is 1.3% of ADV, but the largest trades in thin names (HYG traded only \$3.6M a day in 2008; FXY, DBC and UUP at \$10–22M today) exceed daily volume from about \$100M, so those trades would need to be split across days. Practical capacity: **about \$100M**, set by the currency and commodity ETFs. Using adjusted prices understates early dollar volume, and ETF creation/redemption adds liquidity our ADV measure misses, so both biases make this estimate conservative.

8. Limitations & Next Steps

- **Short out-of-sample period:** 2 years gives a Sharpe standard error of about 0.8; it cannot confirm or reject a 0.4 edge.
- **Selection and hindsight:** Round 2 was designed after seeing Round 1; S3 was chosen among sizings after results; the universe reflects 2026 knowledge.
- **Missing costs:** ETF borrow fees (FXY, USO and EWZ shorts), leverage financing above the T-bill rate, and wider spreads at the open in stress.
- **Data:** Yahoo adjustments rescale history with later dividends (returns unaffected); DTB3 is a discount-basis rate divided by 252.
- **Next:** (1) rerun on 1985+ futures, the original universe, to give S3 statistical power; (2) backtest the class caps as a separately pre-registered trial; (3) a 50/50 blend of trend and long-only, since their crisis returns offset; (4) an informative prior that shrinks each asset's drift toward its asset-class mean, a fuller Bayesian version of S3.

References

- Bailey, D. & López de Prado, M. (2014). The Deflated Sharpe Ratio. *Journal of Portfolio Management* 40(5).
- Barberis, N., Shleifer, A. & Vishny, R. (1998). A Model of Investor Sentiment. *Journal of Financial Economics* 49(3).
- Barroso, P. & Santa-Clara, P. (2015). Momentum Has Its Moments. *Journal of Financial Economics* 116(1).
- Daniel, K. & Moskowitz, T. (2016). Momentum Crashes. *Journal of Financial Economics* 122(2).
- Edwards, W. (1968). Conservatism in Human Information Processing. In *Formal Representation of Human Judgment*, Wiley.
- Grinold, R. & Kahn, R. (2000). *Active Portfolio Management*, 2nd ed. McGraw-Hill.
- Harvey, C., Liu, Y. & Zhu, H. (2016). ...and the Cross-Section of Expected Returns. *Review of Financial Studies* 29(1).
- Hong, H. & Stein, J. (1999). A Unified Theory of Underreaction, Momentum Trading, and Overreaction. *Journal of Finance* 54(6).
- Hurst, B., Ooi, Y. H. & Pedersen, L. (2017). A Century of Evidence on Trend-Following Investing. *Journal of Portfolio Management* 44(1).
- Kaufman, P. (1995). *Smarter Trading*. McGraw-Hill (efficiency ratio).
- Künsch, H. (1989). The Jackknife and the Bootstrap for General Stationary Observations. *Annals of Statistics* 17(3).
- Moreira, A. & Muir, T. (2017). Volatility-Managed Portfolios. *Journal of Finance* 72(4).
- Moskowitz, T., Ooi, Y. H. & Pedersen, L. (2012). Time Series Momentum. *Journal of Financial Economics* 104(2).
- Data:** Yahoo Finance (via the `yfinance` package); Kenneth R. French Data Library (F-F Research Factors and Momentum, daily); Federal Reserve Bank of St. Louis, FRED series DTB3. **Software:** Python, pandas, NumPy, SciPy, statsmodels, matplotlib. AI tools were used for coding assistance; the team reviewed and is responsible for all code and claims.

Appendix (optional): full audit trail

Every hypothesis, decision and result is timestamped in the repository: `HYPOTHESIS.md` (pre-registration), `DECISIONS.md` (choices made after it), and `VARIANTS.md` (every trial, including failures). Commit order: hypothesis 13:44 → data 13:50 → engine (verified on synthetic data) 14:01 → Round 1 results 14:01 → Round 2 pre-registration 20:26 → results 20:27 → S3 selection 20:29 → robustness 20:32 → freeze 20:36 → out-of-sample 20:37 → cash-carry audit 20:50 → corrected reporting 21:04 (all 2026-10-03 ET).

Out-of-sample excess Sharpe, 5 bps	S0	S1	S2	S3	S4
63 days	0.35	0.01	0.03	0.08	−0.10
126 days	0.29	−0.11	−0.11	−0.04	−0.12
252 days	0.84	0.25	0.26	0.21	0.17

Reported for completeness and not used for any choice. The equal-notional S0 at 252 days had the best out-of-sample result (0.84), the reverse of its in-sample rank (0.06). That reversal illustrates how noisy a 2-year window is.